Neuro-Symbolic Reasoning over Patient-Specific Knowledge Graphs for Breast Cancer Progression Prediction

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Computer Science & Engineering

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November, 2026

UNDERTAKING

We declare that the work presented in this report titled "*Neuro-Symbolic Reasoning over Patient-Specific Knowledge Graphs for Breast Cancer Progression Prediction*", submitted to the Computer Science and Engineering Department, Motilal Nehru National Institute of Technology Allahabad, Prayagraj, for the award of the ***Bachelor of Technology*** degree in ***Computer Science & Engineering***, is our original work. We have not plagiarized or submitted the same work for the award of any other degree. In case this undertaking is found incorrect, we accept that our degree may be unconditionally withdrawn.

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CERTIFICATE

Certified that the work contained in the report titled "*Neuro-Symbolic Reasoning over Patient-Specific Knowledge Graphs for Breast Cancer Progression Prediction*", by Shaurya Pratap Singh (20233262), Surya Pratap Singh (20233283), Vishesh Vigh (20233583), and Sehar Majid Abdi (20233561), has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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Abstract

Predicting whether and when a breast cancer will progress after treatment is a central question in oncology, and a prediction helps a clinician only if it comes with reasons. We build a knowledge graph for each of the 1,083 breast cancer patients of the TCGA PanCancer Atlas 2018 cohort. It links the patient to their tumour and to their altered genes, and we join those genes to PrimeKG, a public biomedical knowledge graph of pathways, drugs, and diseases.

Graph neural networks (GNNs) trained on these graphs reach a concordance index (C-index) of about 0.62, and every model family we tried stops near 0.67. We therefore turned the project towards explanation. A survival GNN supplies neural evidence. For the symbolic side we replaced crisp IF–THEN rules with learned logic. We compared four survival-aware reasoners: a Markov logic network whose formula weights are learned from survival times, probabilistic soft logic, rules written by a reinforcement-learning agent, and a hybrid in which reinforcement learning proposes the Markov logic formulas. The hybrid is our rule engine. Each explanation path is scored as $0.6\times$ neural + $0.4\times$ rule.

The logic engine matches RuleKit's accuracy (C-index 0.63 against 0.62). 69% of its weighted formulas reason over knowledge-graph relations, against 10% of RuleKit's rules. The fused model is statistically tied with the best clinical model (0.664 against 0.675). Its month-of-progression estimate is within a year for 48% of progressors. The system writes an explained PDF report for every patient, grouped by when they progressed.

Acknowledgements

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Chapter 1

Introduction and Background

Breast cancer is the most commonly diagnosed cancer worldwide [1]. After treatment, some patients never relapse, some progress within months, and some progress after a decade. Clinicians estimate this risk mainly from stage, lymph-node status, and molecular subtype [2]. Tumour sequencing adds hundreds of gene alterations per patient, but most of them mean little on their own.

A biomedical knowledge graph can put those alterations in context. PrimeKG [3] connects genes to the pathways they act in, the drugs that target them, and the diseases they are linked to. If a patient's altered gene sits in a growth-signalling pathway, the graph can say so explicitly.

1.1 Motivation

A risk score without reasons is hard to trust in a clinic. Neural networks learn well from data but do not explain themselves. Rules are readable but cannot weigh thousands of interacting facts. Neuro-symbolic AI [4] combines the two: a neural part learns from the data, and a symbolic part states its reasoning as explicit, checkable knowledge. Our project began as a GNN benchmark. When accuracy saturated, we followed our supervisor's direction and built a neuro-symbolic layer whose output is a readable report for every patient.

1.2 Problem Statement

Given a patient's clinical record, tumour mutations, and copy-number changes, together with PrimeKG, the system must (i) estimate the risk of progression, (ii) estimate the month of progression when it happens, (iii) explain the estimate with knowledge-graph paths and logic formulas, and (iv) present it as a clinical report grouped by progression time. Outcome data must never leak into the inputs. Every reported prediction must come from a model that did not see that patient during training.

1.3 Contributions and Novelty

1. A leakage-checked pipeline from raw TCGA tables to patient graphs fused with PrimeKG, with 58 unit tests.

2. A benchmark of seven GNN architectures and three training objectives, all evaluated out-of-fold.

3. **Learned logic instead of IF–THEN rules.** We tried all three of our supervisor's suggestions (Markov logic, probabilistic soft logic, and reinforcement learning) as survival reasoners over 422 first-order facts. The chosen engine is new in two ways:

    - it is a *survival* Markov logic network, whose formula weights are learned with the Cox likelihood, so patients who have not relapsed yet are handled correctly;
    - its formulas, many of them multi-step knowledge-graph chains, are *proposed by a reinforcement-learning agent* that is rewarded only for rules that hold up on unseen patients.

4. Explanation paths scored as $0.6\times$ neural importance $+ 0.4\times$ the weight of the formulas they support, with a top-K cutoff.

5. 1,083 patient reports in five progression groups, and two executed notebooks that document the work.

1.4 Background: Data and Knowledge Graph

The Cancer Genome Atlas (TCGA) PanCancer Atlas profiled more than 10,000 tumours with uniform pipelines [5]. We use its breast cancer cohort as distributed by cBioPortal [6]. Its progression-free interval is the recommended outcome for breast cancer [7]. PrimeKG [3] merges 20 biomedical resources, including Reactome pathways and DrugBank targets, into 129,375 nodes and over 8 million edges. Its genes keep their NCBI Entrez identifiers, so patient genes can be matched exactly.

1.5 Background: Graph Neural Networks

A GNN updates every node from its neighbours' features. GCN [8] averages neighbours, GraphSAGE [9] keeps separate self and neighbour weights, attention models such as GATv2 [10] learn a weight per edge, and R-GCN [11] learns one weight matrix per relation type. GNNExplainer [12] finds which edges a trained model relied on for a prediction, which gives every edge an importance score.

1.6 Background: Survival Analysis

Progression is a time-to-event outcome with *censoring*: a patient with no recorded progression may still progress after their last visit. The Kaplan–Meier curve [13] estimates the share of patients still progression-free over time, and the log-rank test [14] compares two such curves. The Cox model [15] turns patient features into a risk score and is fitted by the partial likelihood. Accelerated failure time (AFT) models [16] predict the event time directly. Models are compared with the concordance index (C-index) [17]: the share of patient pairs in which the patient who progressed first received the higher risk. A C-index of 0.5 is a coin flip and 1.0 is perfect.

1.7 Background: Rules and Logic

Survival rules. RuleKit's LR-Rules [18, 19] grows IF–THEN rules such as "IF nodal stage is N0 AND age $\geq$ 41.5 THEN low risk" by maximising the log-rank separation of the patients a rule covers. The rules are crisp: a rule either fires or it does not, and it carries no weight.

Markov logic networks (MLNs) [20] attach a weight to each logic formula. A world that breaks a formula with a large weight becomes less likely but is still possible. Learning an MLN means finding the formulas (structure learning) and their weights.

Probabilistic soft logic (PSL) [21] allows truth values between 0 and 1 and turns each rule into a penalty for how far it is from being satisfied. Inference is then a convex optimisation, which allows *collective* reasoning, for example letting similar patients influence each other.

Reinforcement learning (RL) for rules. A policy-gradient agent [22] learns from rewards rather than labels. In knowledge-graph reasoning, agents have been trained to walk relation paths [23, 24] and to write logic rules [25]. We use RL to write survival rules.

Chapter 2

Proposed Work

Figure 1 gives the overview. Patient graphs are built from TCGA and fused with PrimeKG. A survival GNN and the logic reasoners run on the same patients and the same five folds. Their risks are combined in one Cox model, and every patient gets a report that explains the prediction with knowledge-graph paths and logic formulas.

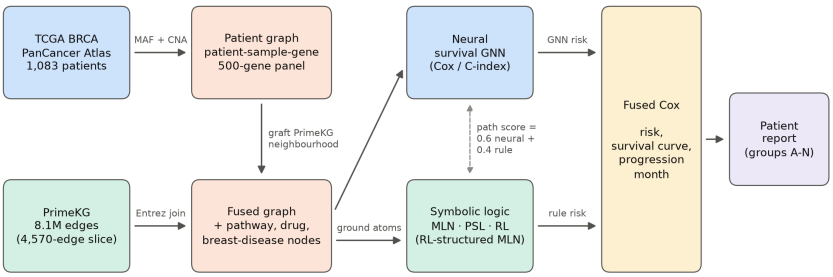


Figure 1: Overview of the neuro-symbolic progression system.

2.1 Data and Labels

We use the cBioPortal `brca_tcga_pan_can_atlas_2018` study, checked against its official checksums: the clinical tables, 126,252 mutation records, and copy-number calls for 25,128 genes. 1,083 patients have a usable progression-free survival (PFS) record, and 145 of them progressed. Outcome columns (survival, progression, and

tumour-status fields) are blocked from every feature table by a guard function that the tests check.

The GNN benchmark uses a 36-month landmark label (498 patients, 89 events). The survival models and the reports use the full follow-up time and event of all 1,083 patients.

2.2 Patient Graphs and PrimeKG Fusion

Each patient graph has the shape Patient → Sample → Gene. Genes are limited to a 500-gene panel of the most frequently altered genes plus 27 known breast cancer drivers. PrimeKG's 8.1 million edges are filtered once to 4,570 edges that touch the panel: protein–protein, gene–pathway, gene–drug, and gene–breast-disease links. A patient's genes join PrimeKG on their Entrez identifiers, which adds each altered gene's partners, pathways, drugs, and breast-disease links (Figure 2). The median fused graph has 632 nodes.

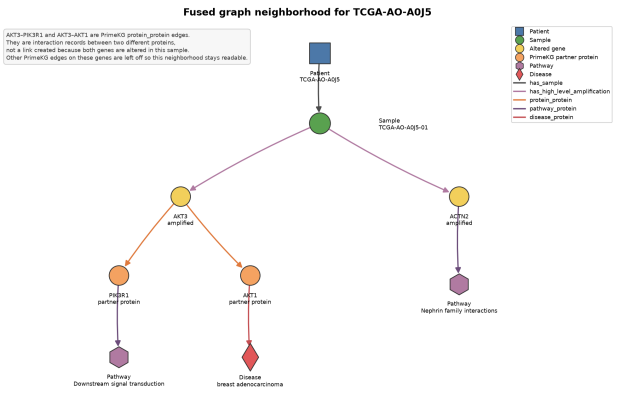


Figure 2: Part of a fused graph: a patient's altered genes linked to PrimeKG proteins, pathways, and a breast-disease node.

2.3 Neural Part: Graph Neural Networks

All GNNs share three message-passing layers of width 64, a readout over the patient and altered-gene nodes, and a clinical shortcut. On the landmark endpoint they are

trained with cross-entropy or a smooth C-index loss. On the full follow-up they are trained with the Cox partial likelihood, as in DeepSurv [26]. The best survival GNN supplies a neural risk. GNNExplainer, run on the fold model that held the patient out, gives every edge of the patient's graph an importance score.

2.4 Symbolic Part: From IF–THEN Rules to Learned Logic

Baseline. RuleKit LR-Rules over 188 readable predicates gives about 8 crisp rules per fold. Only 10% of them mention the knowledge graph, and none of them carry a weight.

Ground atoms. All new reasoners read the same 422 facts about each patient. There are 26 clinical atoms such as $\mathit{NStage}(p, \mathrm{N3})$, 57 genomic atoms such as $\mathit{Mutated}(p, \mathrm{TP53})$, and 339 knowledge-graph atoms such as

$$\mathit{Altered}(p, g) \wedge \mathit{InPathway}(g, \text{PI3K Cascade}),$$

whose value is the number of the patient's altered genes in that pathway. Atoms are chosen by prevalence (3–90% of patients), never by outcome.

1. Survival Markov logic network. Every formula has the form $w_k : \mathit{Body}_k(p) \Rightarrow \mathit{Progress}(p)$. A body is one atom or a clinical × knowledge-graph pair found on the training fold. A patient's risk is the weighted count of the formulas that hold for them:

$$\eta(p) = \sum_k w_k \log\big(1 + n_k(p)\big), \tag{1}$$

where $n_k(p)$ is the number of ways formula k is true for patient p. The novelty is in how w is learned. A standard MLN would treat every patient who has not relapsed *yet* as a non-progressor. We instead learn w with the Cox partial likelihood, which uses each patient's follow-up time correctly, plus an elastic-net penalty [27]. The penalty sets the weight of useless formulas to exactly zero, which performs structure learning.

2. Survival PSL. Truth values are soft. The program has a rule per fact ("fact ⇒ high risk" and "fact ⇒ low risk") and a collective rule, $Similar(p, q) \wedge HighRisk(q) \Rightarrow HighRisk(p)$, that links each patient to their ten most similar training patients. Censoring is handled with soft targets: a patient censored at 20 months gets the Kaplan–Meier probability of progressing before 36 months, given that they were still progression-free at 20.

3. Reinforcement-learned rules. A small recurrent network writes a rule one atom at a time (up to three atoms) and is trained with REINFORCE [22]. Its reward measures *stability*. The training fold is split into two halves, and a rule is rewarded only if it separates fast from slow progressors in the same direction in both halves. A rule is kept only if it also separates them on a further 30% of the fold that the search never saw.

4. RL-structured survival MLN (the engine). The rules that survive RL's filter are added to the MLN's candidate formulas, and the survival MLN learns every weight. RL does the structure learning and the survival likelihood does the weight learning. Reasoners within one standard error of the best mean C-index are treated as tied [28], and among them the one with the richest reasoning becomes the engine.

2.5 Fusion, Model Choice, and Month of Progression

The final risk comes from a Cox model on three inputs: the 19 clinical fields, the GNN risk, and the logic-engine risk. It is compared with clinical-only Cox models and a random survival forest [29]. Among the models tied with the best, we choose the one that can explain its prediction.

Most patients never progress, so we estimate the month of progression *given* that progression happens. A log-logistic AFT model is fitted on the training-fold progressors and checked against the recorded months of held-out progressors.

2.6 Explanation Paths: 0.6 Neural + 0.4 Rule

A candidate path is a real walk in the patient's fused graph: patient → sample → altered gene → (partner protein) → pathway, drug, or breast disease. Its score combines the two halves of the system:

$$\text{score}(\text{path}) = 0.6 \times \text{neural} + 0.4 \times \text{rule}. \tag{2}$$

Neural is the percentile of the path's mean GNNExplainer edge importance. *Rule* is the summed weight of the engine's formulas that the path supports, scaled to the range 0 to 1. A formula counts when it holds for the patient and mentions the path's end node, its altered gene, or a gene on the path. Its sign shows whether the evidence points to faster or slower progression. A path is *prominent* if its score is in the top 20% of the cohort. A report shows $K = \min(5, \text{prominent paths})$ paths and tops up to at least three.

2.7 Patient Reports

Reports are grouped by the recorded progression month: A ($\leq$12 months, 32 patients), B (12–24, 30), C (24–36, 27), D (>36, 56), and N (no progression recorded, 938). Each report is a two- to three-page PDF. It shows the clinical record, the risk percentile and the chance of progression at 12, 24, 36, and 60 months, the estimated progression month next to the recorded outcome, and the patient's survival curve. It then lists the top-K paths, the weighted formulas that fire for the patient, the molecular profile, and the most similar patients. Each group also has a summary page.

Chapter 3

Results and Analysis

3.1 Setup

All experiments ran on an 8-core laptop without a GPU, in Python with PyTorch Geometric [30], lifelines [31], scikit-survival [32], and RuleKit. Every result comes from stratified five-fold cross-validation with a fixed seed. Every prediction is *out-of-fold*: it comes from models trained on the four folds that exclude that patient. The full pipeline takes about 50 minutes, and the 1,083 reports take about 5 minutes.

3.2 Graph Neural Networks: Accuracy Saturates

Table 1 summarises the GNN benchmark on the 36-month landmark. The clinical logistic regression is the strongest model. GCN comes close, and the gene-only network is weakest. Relation-aware models, a C-index loss, and graph structure learning [33], which learns which PrimeKG edges to keep, all stayed within the fold-to-fold noise. On an external microarray cohort (GSE2034 [34]) the graph models also did not beat a linear baseline.

3.3 Symbolic Reasoners Compared

Table 2 and Figure 3 compare the four reasoners with RuleKit on the full follow-up.

Table 1: 36-month progression (498 patients, 89 events), five-fold mean ± standard deviation. A random ranking has an AUPRC of 0.18.

Model	AUROC	AUPRC
Clinical logistic regression	0.703 ± 0.081	0.463 ± 0.133
GCN	0.680 ± 0.076	0.396 ± 0.091
GraphSAGE	0.650 ± 0.045	0.379 ± 0.091
GATv2	0.631 ± 0.046	0.355 ± 0.060
MLP (clinical + gene list)	0.562 ± 0.098	0.262 ± 0.064

Table 2: Symbolic reasoners: five-fold C-index, formulas kept per fold, and the share of formulas that use PrimeKG.

Reasoner	C-index	Formulas/fold	Use PrimeKG
Survival MLN	0.630 ± 0.055	32.8	63%
RL-structured MLN (engine)	$\mathbf{0.628 \pm 0.055}$	35.2	**69%**
RuleKit IF–THEN rules (baseline)	0.619 ± 0.048	7.8	10%
Survival PSL	0.604 ± 0.059	40.0	63%
RL rules alone	0.563 ± 0.046	25.6	91%

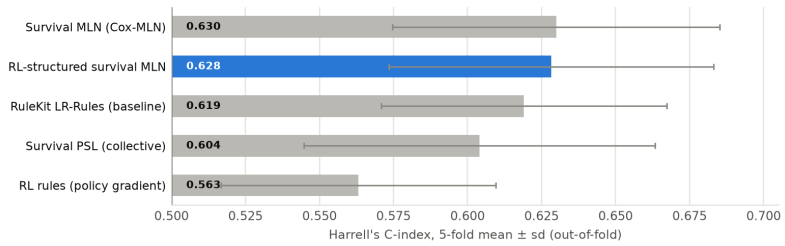


Figure 3: Mean C-index of the symbolic reasoners over five folds. The chosen engine is highlighted.

What the comparison shows.

1. **Accuracy is a tie.** Both MLN variants lead, but their gap over RuleKit (about 0.01) is smaller than the swing between folds (about 0.05). MLN, RL-structured MLN, and RuleKit are statistically tied, and the preference order selects the RL-structured MLN.

2. **The reasoning is much richer.** The engine keeps about 35 weighted formulas per fold instead of 8 crisp rules, and 69% of them reason over PrimeKG, about seven times RuleKit's share. Over the five folds it kept 176 formulas: 51 single atoms, 81 mined clinical × PrimeKG pairs, and 44 multi-step formulas written by RL.

3. **Collective reasoning found nothing.** PSL learned a weight of about 0.01 for "similar patients have similar risk". In this cohort, a patient's neighbours add no information beyond the patient's own facts.

4. **RL alone overfits.** The agent tried 20,000–27,000 different rules per fold, so its best-looking rules are partly luck. RL works well as the formula *proposer* for the MLN, where the survival likelihood decides which proposals count.

Examples of learned formulas. Clinical facts still carry the largest weights, for example $\mathit{NStage}(p, \mathrm{N0}) \Rightarrow \mathit{Progress}(p)$ with weight -0.30 (slower progression). A mined formula raises risk (+0.15, 43 patients):

$$\mathit{NStage}(p, \mathrm{N3}) \wedge \mathit{Altered}(p, g) \\ \wedge \mathit{LinkedTo}(g, \text{breast ductal adenocarcinoma}) \Rightarrow \mathit{Progress}(p).$$

An RL-proposed two-pathway formula is protective (-0.14, 93 patients; Figure 4):

$$\mathit{Altered}(p, g) \wedge \mathit{InPathway}(g, \text{VEGFA-VEGFR2}) \\ \wedge \mathit{Altered}(p, g') \wedge \mathit{InPathway}(g', \text{Ion homeostasis}) \Rightarrow \mathit{Progress}(p).$$

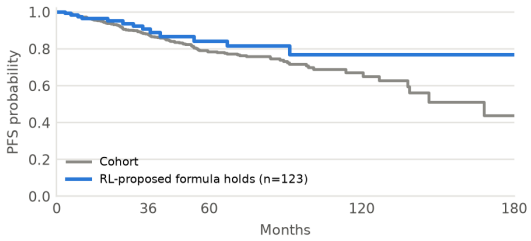


Figure 4: Kaplan–Meier curve of the patients for whom the RL-proposed VEGFA-VEGFR2 / ion-homeostasis formula holds, against the whole cohort.

3.4 The Best Survival Model

Table 3 compares the main models on the full follow-up. The best mean C-index is 0.675 (clinical Cox), with one standard error of 0.016. Every model above 0.659 is statistically tied with it. Among the tied models we chose the **neuro-symbolic Cox model** (0.664) because it is the only one whose prediction can be explained with paths and formulas.

Table 3: Full follow-up (1,083 patients, 145 events): five-fold C-index and integrated Brier score to 120 months (IBS, lower is better).

Model	C-index	IBS
Clinical Cox (19 fields)	0.675 ± 0.035	0.134
Neuro-symbolic Cox (clinical + GNN + logic)	$\mathbf{0.664 \pm 0.021}$	0.134
GNN + logic, no clinical	0.642 ± 0.056	0.139
RL-structured MLN alone	0.628 ± 0.055	–
Best GNN (GraphSAGE, C-index loss)	0.622 ± 0.067	–
Random survival forest	0.554 ± 0.023	0.145

Two findings matter. First, every model family (GNN, logic, forest, and Cox) stops near a C-index of 0.67. With 145 events, stage, nodal status, age, and subtype carry most of the signal. Second, without clinical inputs the GNN and logic risks together reach 0.642, better than either alone, but once the clinical fields are present they add no independent information. The neuro-symbolic layer therefore adds *explanation, not accuracy.* We state this plainly in the report and in every patient report.

3.5 Month of Progression

On the 145 held-out progressors, the log-logistic AFT model is within 12 months of the recorded month for 48% of patients, against 37% for a constant guess. It places 33% of them in the correct 12-month window, against 19% for the constant guess and 25% by chance. Its median error is 12.9 months (Figure 5). Estimates are pulled towards 15–45 months: very early progressions are overestimated and very late ones underestimated.

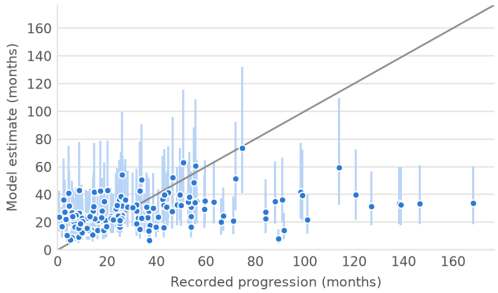


Figure 5: Estimated against recorded progression month for held-out progressors. The diagonal is perfect agreement.

3.6 Explanation Paths and Reports

The system scored 59,598 candidate paths for 1,033 patients. The other 50 patients have no altered gene with a PrimeKG link. The prominence cutoff is a score of 0.523. 35% of the paths shown in reports are backed by at least one learned formula, and a median of three formulas fires per patient. Figure 6 shows the top paths of patient TCGA-BH-A1EV (group A, progression at 12.0 months). All four are backed by formulas that raise the hazard. The strongest runs from the amplified gene NFASC to the pathway *Neurofascin interactions*. Figure 7 shows the page of the patient's report that lists these paths and the formulas that fire.

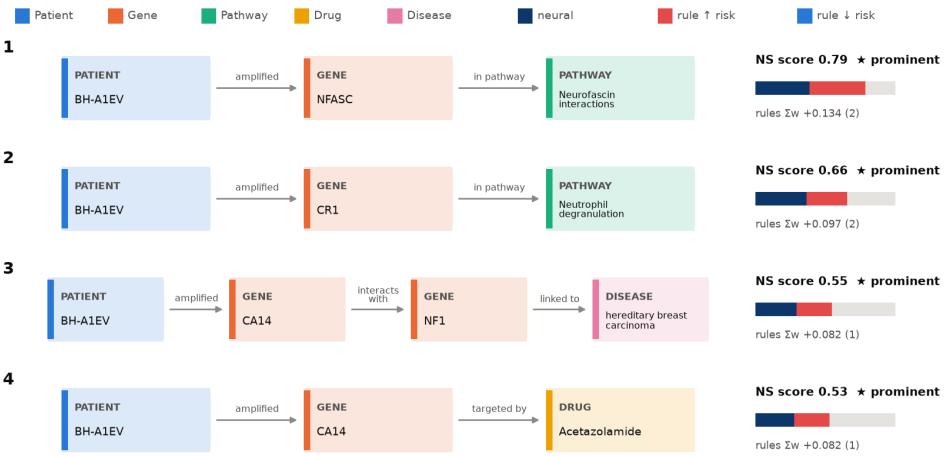


Figure 6: Top-K paths of TCGA-BH-A1EV. The bar on the right splits each score into its neural part (navy) and its rule part (red: the formulas raise the hazard).

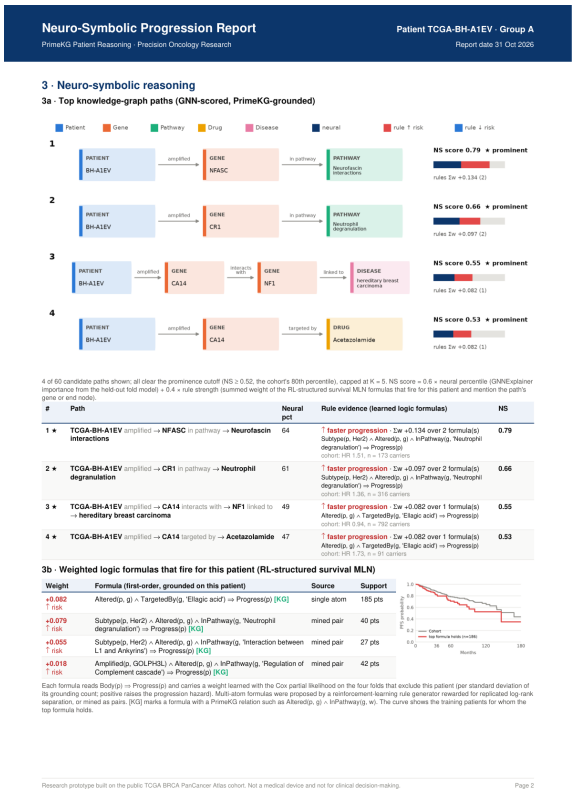


Figure 7: Page 2 of the report of TCGA-BH-A1EV: top-K knowledge-graph paths and the weighted logic formulas that fire.

Chapter 4

Conclusion and Future Work

4.1 Conclusion

We built a pipeline from raw TCGA tables to patient knowledge graphs fused with PrimeKG and benchmarked graph neural networks on them. We then turned the system into a neuro-symbolic progression model with a report for every patient. Following our supervisor's suggestion, we replaced IF–THEN rule induction with learned logic and compared Markov logic, probabilistic soft logic, and reinforcement learning. The chosen engine is a survival Markov logic network: its formulas are proposed by a reinforcement-learning agent and weighted with the Cox likelihood.

The engine matches the accuracy of the old rules and says far more. It keeps about four times as many formulas, they carry weights, and 69% of them reason over PrimeKG instead of 10%. These formulas now drive the 0.4 rule part of every explanation path. The fused model is tied with the best clinical model (C-index 0.664 against 0.675), and every prediction is out-of-fold and leakage-checked. On this cohort, knowledge-graph reasoning adds explanation rather than accuracy, and the system is designed around that finding.

4.2 Limitations

- Accuracy saturates at a C-index of about 0.67 with 145 events.

- The month estimate is coarse, and late progressions are hard to place.
- Paths show what the model relied on. They are not causal claims, and PrimeKG drug links are not treatment advice.
- The work uses one cohort and one random seed, and the reports have not been validated externally.

4.3 Future Work

- Let the RL agent walk multi-step PrimeKG relation paths and train it jointly with the MLN weights.
- Add expression and methylation data, and use pretrained knowledge-graph embeddings.
- Validate the model and the reports on an external cohort such as METABRIC.
- Ask oncologists to rate how useful and correct the reports are.

The code, the commands that reproduce every result, and two executed notebooks are in the project repository: `notebooks/neurosymbolic_patient_reports.ipynb` walks through the whole system, and `notebooks/symbolic_reasoning_novelty.ipynb` explains the logic reasoners and the novelty points.

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